

01 Execution Evidence Report

Step-by-step build evidence, screenshots, validation, decisions, and ownership disclosure.

Repository	https://github.com/aalanani2000/al-rouf-ai-integration-assessment
Default branch	main
Latest commit	653ec552871083dca14498d120d6e6c156ed4f31
Prepared date	2026-06-24
Assessment	AI Integration Engineer task pack

Assessment Requirements Covered

- Runnable GitHub repository with setup documentation, task folders, screenshots, and report deliverables.
- RFQ to CRM automation using n8n as an equivalent orchestration approach.
- FastAPI quotation microservice with tests, Docker packaging, and OpenAPI documentation.
- Bilingual RAG workflow over English and Arabic sample documents with citations and refusal behavior.
- Explicit notes on error handling, maintainability, security hygiene, trade-offs, validation, and AI assistance.

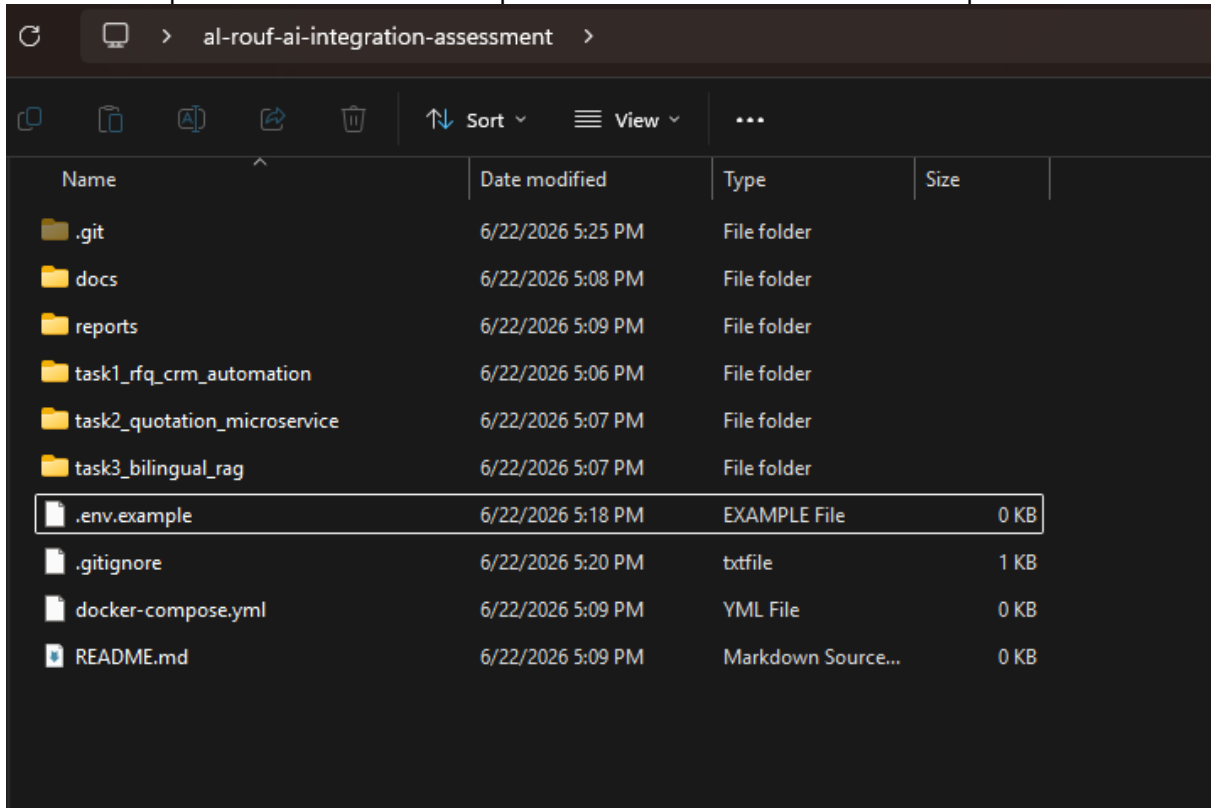
Work Evidence By Task

Task	Implementation	Validation	Status
Task 1	n8n RFQ webhook, OpenAI extraction, mock CRM, bilingual reply, alert log.	Successful n8n executions, persisted JSON records, screenshot evidence.	Complete; attachment bytes scoped as future enhancement.
Task 2	FastAPI quotation API with catalog, pricing tiers, SQLite, Docker, OpenAPI.	Automated pytest suite and OpenAPI screenshot evidence.	Complete.
Task 3	FastAPI bilingual RAG service over English/Arabic documents with citations and refusal.	Unit tests, Arabic question evidence, error-handling screenshot.	Complete.

Task 1 Evidence: RFQ To CRM Automation

- 1 Created a Docker-based n8n environment with persistent volumes.
- 2 Built a webhook intake endpoint for RFQ-style JSON payloads.
- 3 Configured OpenAI extraction for urgency, key requirements, and summary.
- 4 Merged structured fields with extracted intent and generated CRM metadata.
- 5 Persisted CRM records and sales alerts with bilingual reply drafts.
- 6 Debugged sandbox, missing-file, and node-reference issues during testing.

- Trade-off: webhook simulation kept the workflow reproducible without OAuth setup.
- Limitation: attachment filenames are captured as metadata; real file storage is a future enhancement.
- Validation: multiple full workflow executions produced successful CRM and alert outputs.



Task 1 evidence: repository structure.png

```
PS C:\Users\idhoo\Desktop\al-rouf-ai-integration-assessment> docker inspect n8n --format '{{ .Mounts }}'
[{"volume": "n8n_data", "n8n_data": "/home/node/.n8n", "local": "rw", "true": true}, {"bind": "C:\\Users\\idhoo\\Desktop\\al-rouf-ai-integration-assessment\\data", "data": "/data", "rw": "true", "private": true}]
```

Task 1 evidence: proof of the volumes being correctly attached.png

Webhook Listen for test event

Parameters Settings Docs

Webhook URLs

Test URL Production URL

POST http://localhost:5678/webhook-test/rfq-intake

HTTP Method

POST

Path

/rfq-intake

Authentication

None

Respond

When Last Node Finishes

Response Data

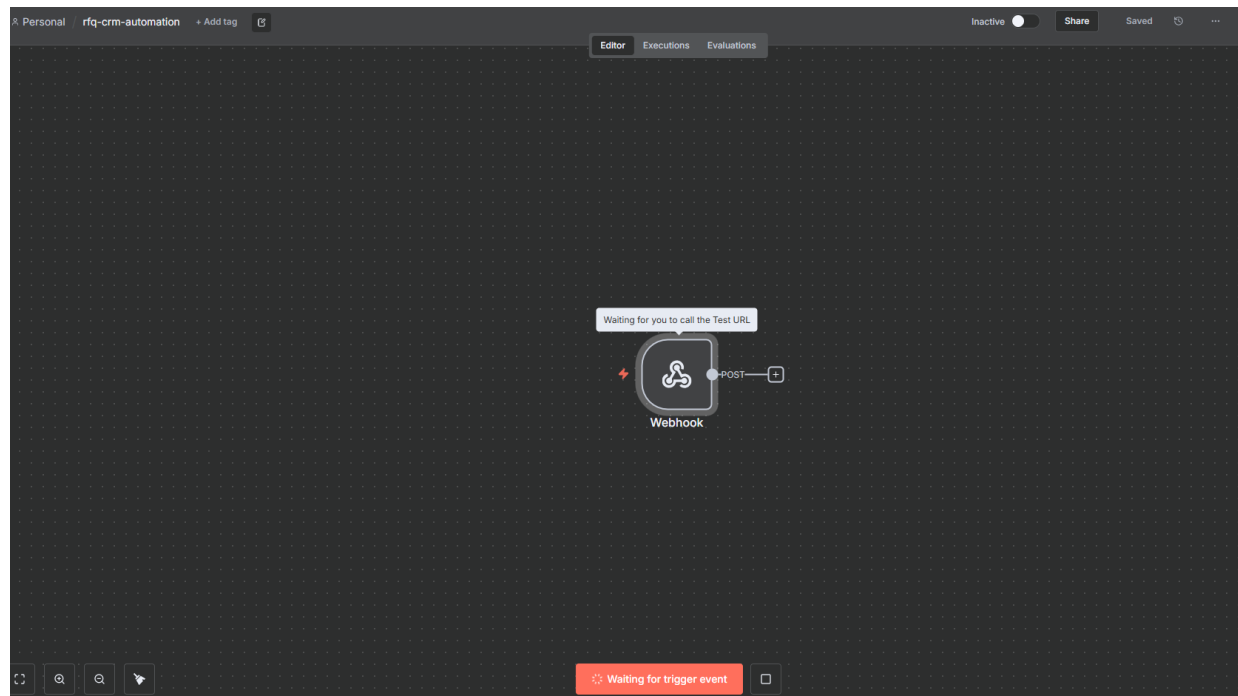
First Entry JSON

Options

No properties

Add option

Task 1 evidence: Webhook node.png



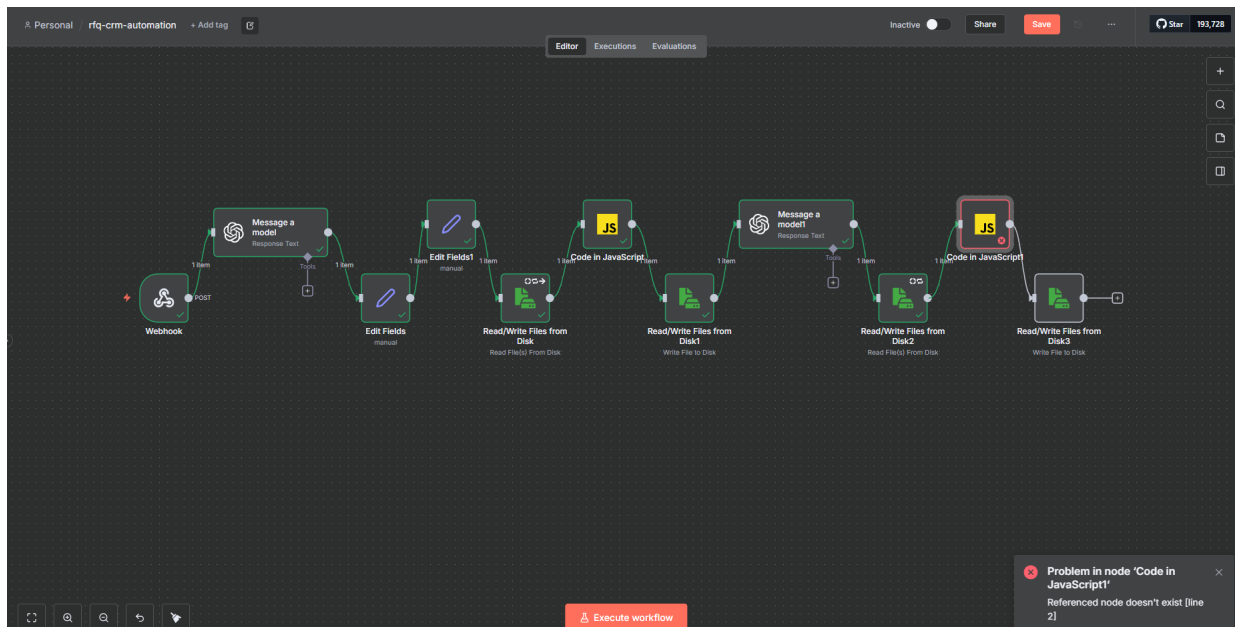
Task 1 evidence: n8n canva.png

OUTPUT 🔍 Schema Table JSON 🔗

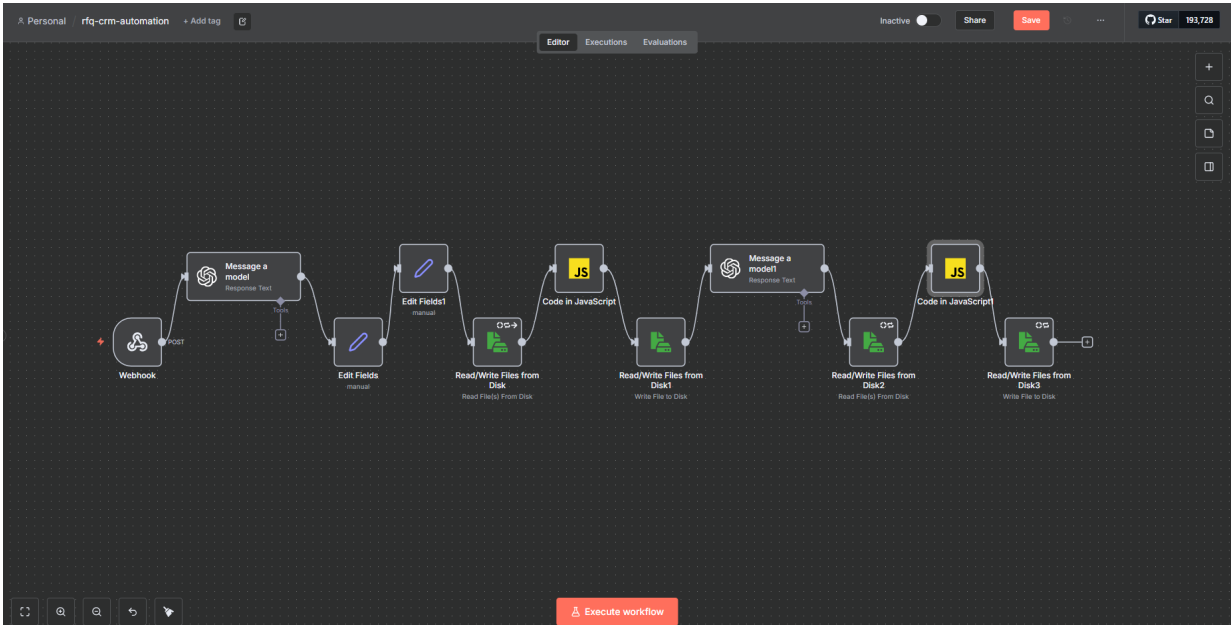
1 item

T	company_name	Gulf Construction LLC
T	contact_name	Ahmed Al-Mutairi
T	contact_email	ahmed@gulfconstruction.sa
T	contact_phone	+966501234567
T	product_interest	LED High Bay Lights, 150W
T	quantity	200
T	project_location	Jeddah, Saudi Arabia
T	deadline	2026-07-15
T	attachments	["site_plan.pdf","spec_sheet.pdf"]
T	extracted_intent	{"urgency":"standard","key_requirements":["include delivery timeline","bulk pricing for 200 units"],"summary":"Requesting a quotation for 200 units of 150W LED high bay lights for a warehouse project in Jeddah, with delivery timeline and bulk pricing included."}

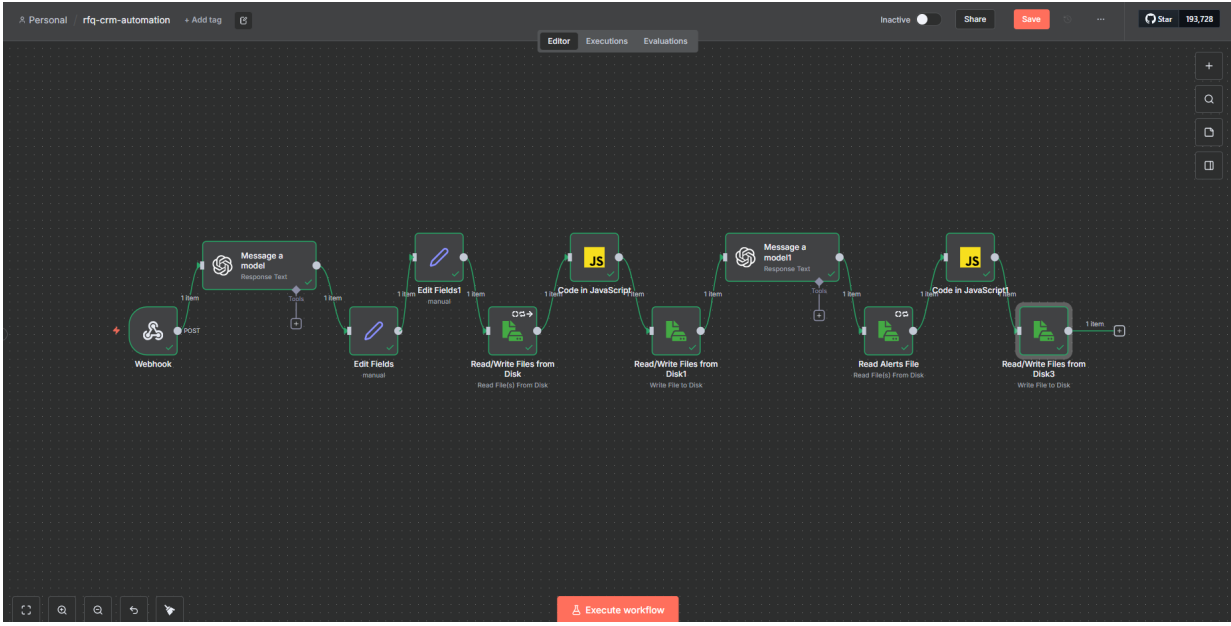
Task 1 evidence: set field output.png



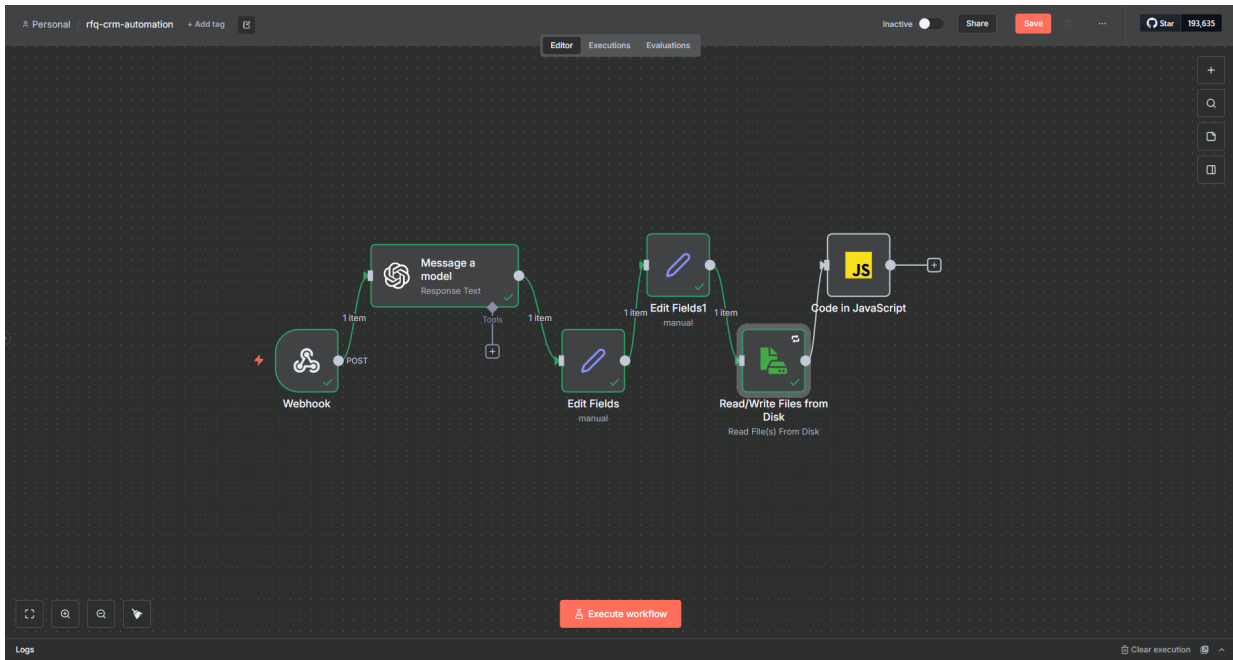
Task 1 evidence: javascript code 1 error .png



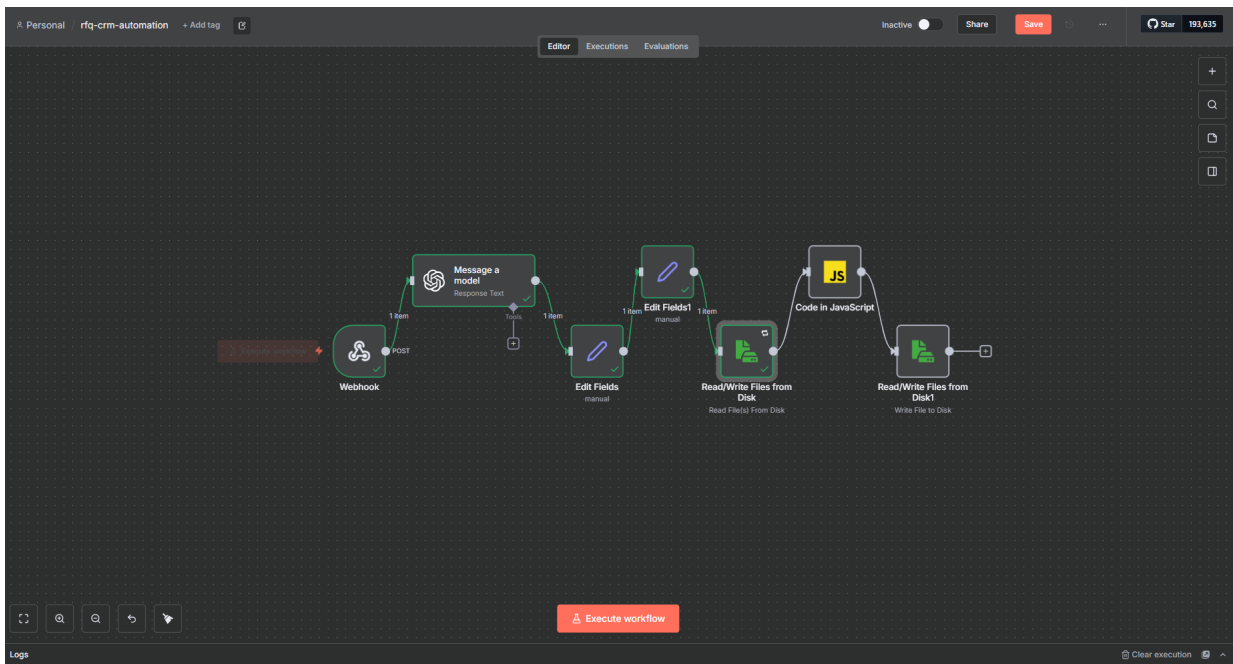
Task 1 evidence: workflow progress .png



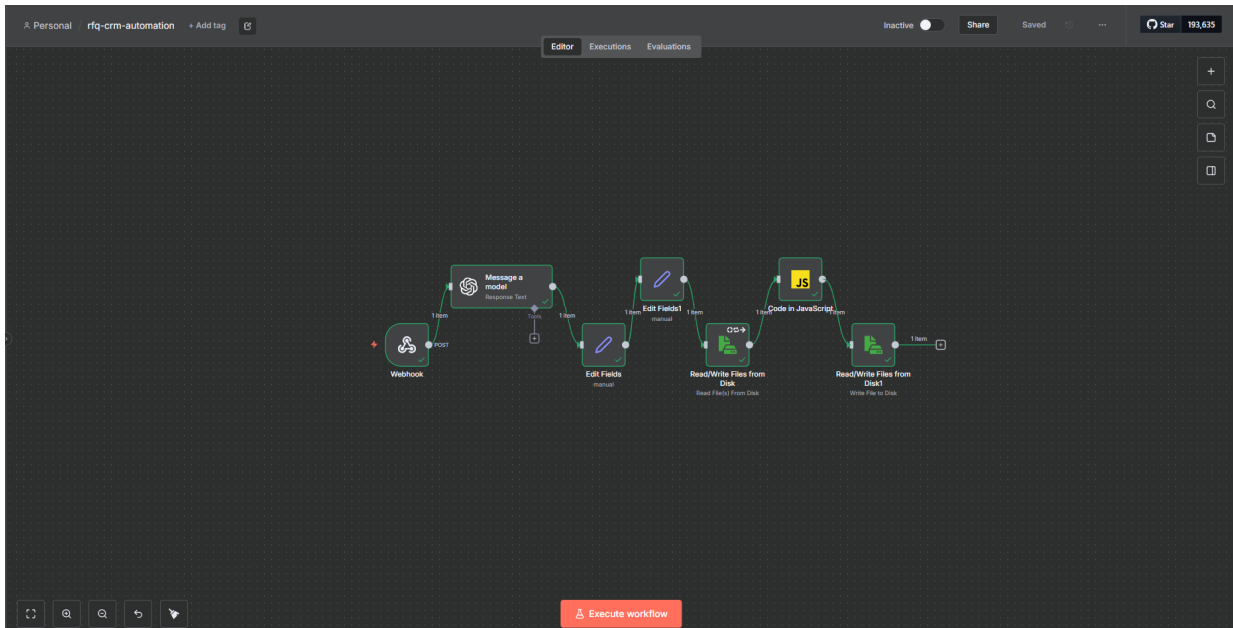
Task 1 evidence: workflow progress 1 .png



Task 1 evidence: workflow.png



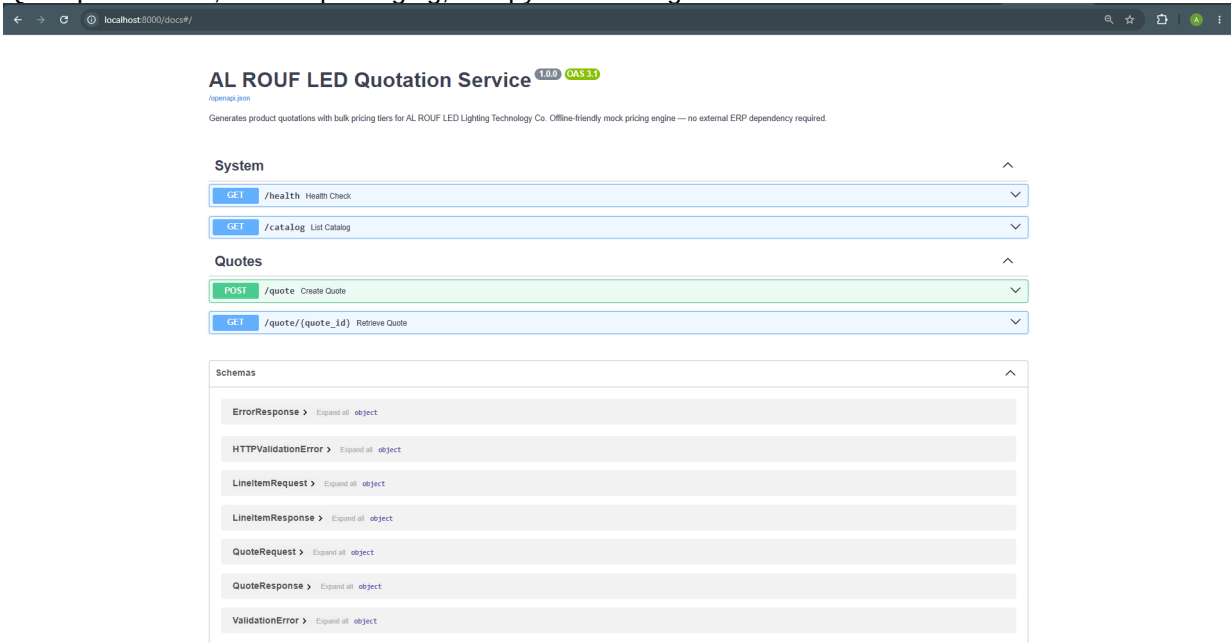
Task 1 evidence: workflow2.png



Task 1 evidence: workflow succes.png

Task 2 Evidence: Quotation Microservice

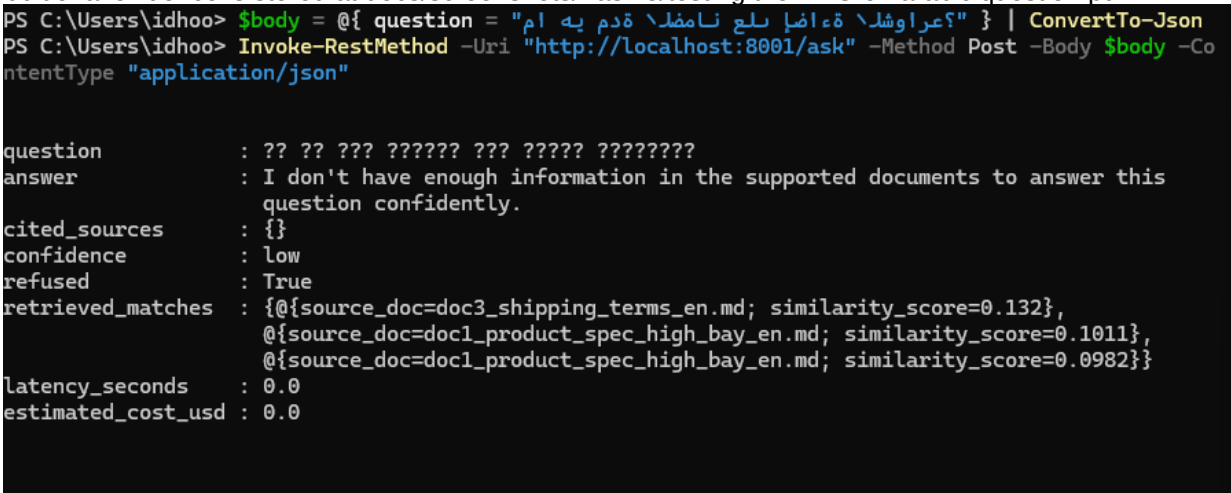
- FastAPI service with health, catalog, quote creation, and quote retrieval endpoints.
- Product catalog and bulk-discount logic with offline-friendly mocked data.
- SQLite persistence, Docker packaging, and pytest coverage.



Task 2 evidence: OpenAPI documentation page.

Task 3 Evidence: Bilingual RAG Knowledge Workflow

- FastAPI service loads 4 sample documents in English and Arabic.
- Embeddings are cached locally and searched with cosine similarity.
- The service returns cited answers and refuses out-of-scope questions.
- Additional evidence is stored at docs/screenshots/Task 3/testing the RAG on arabic question.pdf.



Task 3 evidence: model communication issue captured during testing.

Validation Summary

- Task 1 was validated through end-to-end n8n executions and persisted mock output files.
- Task 2 includes a pytest suite and OpenAPI evidence.
- Task 3 includes offline tests plus optional API-key integration tests.
- Docker packaging is included for the runtime services.

Ownership And AI Assistance Disclosure

AI assistance was used as a technical mentor and documentation partner for architecture discussion, starter logic, debugging support, and final writing polish. The candidate personally completed the hands-on implementation work: n8n node setup, Docker execution, API implementation, test execution, screenshot collection, environment configuration, validation, and final engineering decisions.